

Size and Asymptotics

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July 2026

Number theory is often about finding integer solutions to equations of the forms $A(p, q, r, \dots) = B(p, q, r, \dots)$ or $A|B$, where A, B are functions in the parameters $p, q, r, \dots$. The main idea of today is that oftentimes, an easy way to approach this is to determine when $A > B$.

We will be working with two different aspects of this size idea:

1. Oftentimes the fact that we are using *integer* data can give some information on the size of certain functions. For example, we know that the proper divisors of an integer n are at most $n/2$.
2. Sometimes it is also useful to consider how our functions grow as our parameters go to infinity. For example, if A increases much more rapidly than B , then $A = B$ cannot occur very often.

Disclaimer: This handout is similar (but not identical) to the D Size and D EuclidAlg handouts from OTIS. The writing is my own though I took some inspiration from these two handouts. If you want more practice for problems on today's topics and you have access to OTIS, I would recommend looking at these handouts.

1 Integer Divisibility

Our two types of integer equations, $A|B$ and $A = B$, are quite interrelated since we can often factor integer expressions. Here are some important properties that are often useful when dealing with integer divisibilities:

1. The divisors of an integer n are usually much smaller than n , since they must be of the form n/k for some integer k .
2. Oftentimes it's useful to factor whatever polynomial expressions you have, especially because oftentimes, the factored expressions have to be relatively prime to each other. For instance, given the equation $4x^2 - 1 = y^5$, we may factor $4x^2 - 1 = (2x - 1)(2x + 1)$, and note that $\gcd(2x - 1, 2x + 1) = 1$. Thus $2x - 1$ and $2x + 1$ must both be 5th powers as well.
3. In a similar vein: given a polynomial divisibility $P(n)|Q(n)$, it's often useful to divide P out from Q using the *Euclidean Algorithm*, to express this divisibility as $P(n)|R(n)$, where R is the remainder polynomial.
 - (a) An especially useful identity in this vein is that $n|P(a + n) - P(a)$ for any (nonzero) integers a, n and integer polynomial P .

Here, recall that the Euclidean Algorithm is the following idea. Given any two (integers / polynomials) A, B , we have that $\gcd(A, B) = \gcd(A, B - A)$. So we can repeatedly subtract one entry out of the other to compute the gcd.

Here's some practice with integer divisibility.

Example 1.1. [APMO 2013 2] Determine all positive integers n for which $\frac{n^2 + 1}{[\sqrt{n}]^2 + 2}$ is an integer. Here $[r]$ denotes the greatest integer less than or equal to r .

1. Before we begin, estimate how large $(n^2 + 1)/([\sqrt{n}]^2 + 2)$ is in terms of n . What does this tell you about what integer values this expression can take?
2. Let's get to bounding. Write $m = \lfloor \sqrt{n} \rfloor$, so $n = m^2 + p$ for $p \in [0, 2m]$. Then calculate the polynomial remainder of $\sqrt{(m^2 + p)^2 + 1}m^2 + 2$, and bound how large it can be.
3. Tackle each possible value of the remainder and solve the problem.

Many NT functional equations are susceptible to the integer divisibility ideas above.

Example 1.2. [ISL 2016 N6] Denote by $\mathbb{N}$ the set of all positive integers. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for all positive integers m and n , the integer $f(m) + f(n) - mn$ is nonzero and divides $mf(m) + nf(n)$.

1. What's your guess for the solution set?
2. Let's start with the $m = n$ case. Show that $2f(m) - m^2 \mid m^3$.
3. Determine $f(1)$. Also show that for all primes p , $f(p)$ takes one of 6 values.
4. Pin down $f(p)$ for all sufficiently large p .
5. Finish the problem.

1.1 Integer Divisibility / Euclid Algorithm Problems

1. $\diamond$ Let $\mathbb{Z}^+$ be the set of positive integers. Determine all functions $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ such that $a^2 + f(a)f(b)$ is divisible by $f(a) + b$ for all positive integers a, b .
2. $\diamond$ Find all pairs (a, b) of positive integers such that a^3 is multiple of b^2 and $b - 1$ is multiple of $a - 1$.
3. $\star$ Let a, b be positive integers such that $a + b^3$ is divisible by $a^2 + 3ab + 3b^2 - 1$. Prove that $a^2 + 3ab + 3b^2 - 1$ is divisible by the cube of an integer greater than 1.
4. $\diamond$ Let $\mathbb{Z}_{>0}$ be the set of positive integers. Find all functions $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ such that

$$m^2 + f(n) \mid mf(m) + n$$

for all positive integers m and n .

5. Find all positive integers $n \geq 1$ such that there exists a pair (a, b) of positive integers, such that $a^2 + b + 3$ is not divisible by the cube of any prime, and

$$n = \frac{ab + 3b + 8}{a^2 + b + 3}.$$

6. $\diamond$ Let $a_1, a_2, \dots, a_n$ be distinct positive integers, $n \geq 3$. Prove that there exist distinct indices i and j such that $a_i + a_j$ does not divide any of the numbers $3a_1, 3a_2, \dots, 3a_n$.
7. $\diamond$ Find all functions $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ such that $a + f(b)$ divides $a^2 + bf(a)$ for all positive integers a and b with $a + b > 2019$.
8. $\star$ Find all pairs of positive integers (a, b) satisfying the following conditions: a divides $b^4 + 1$, b divides $a^4 + 1$, $\lfloor \sqrt{a} \rfloor = \lfloor \sqrt{b} \rfloor$.
9. Let f be a function from the set of integers to the set of positive integers. Suppose that, for any two integers m and n , the difference $f(m) - f(n)$ is divisible by $f(m - n)$. Prove that, for all integers m and n with $f(m) \leq f(n)$, the number $f(n)$ is divisible by $f(m)$.
10. $\diamond$ Let a, b be positive integers such that $b^n + n$ is a multiple of $a^n + n$ for all positive integers n . Prove that $a = b$.

2 Classical Size Problems

Remember that the above integer divisibility techniques are simply tools that let us interpret divisibility conditions. The ultimate goal of today's lesson is still to adopt the habit of viewing *equalities* through the lens of *inequalities*. In more general applications, you may need to do some more refined inequality estimates to get the bounds you want.

Let's start with a prototypical example of a size problem.

Example 2.1. [ISL 2019 N2] Find all triples (a, b, c) of positive integers such that $a^3 + b^3 + c^3 = (abc)^2$.

1. Let's do some simple mental estimates. If a, b, c are all large and around the same size, is this equation possible? What does this tell you about the relative sizes of a, b, c ?
2. Let's now set $a \geq b \geq c$. Can you bound how large b, c are compared to a ?
3. Now write the equation as $a^2((bc)^2 - a) = b^3 + c^3$. Using this, derive a bound on b and c , and show $c = 1$.
4. With this information, improve your earlier inequalities to show $b \leq 4$. Solve the problem from here.

The first step here is the most important. This equation shouldn't be true "generically", so equalities have to occur in wonky conditions, which you can bound to retrieve the answer.

Here's a more involved example involving factoring.

Example 2.2. [Canada Winter Buffet 2019 N2] Are there 7 distinct odd prime numbers satisfying that the difference of the eighth powers of any two of them is divisible by the remaining 5 primes?

Please try this problem for a bit before looking at the walkthrough.

1. This problem has a lot of data, and it's not clear what to work with. Replace 7 with n to start; then we want to examine when divisibilities of the form $p_1 p_2 \dots p_{n-2} | p_{n-1}^8 - p_n^8$ are possible, where the condition gives us some choice on how p_{n-1}, p_n relate to the other primes. Play around a bit to figure out which primes p_{n-1}, p_n we want to choose.
2. Can you solve the problem for $n = 10$?
3. We'll have to try a little bit harder for seven. Remember that we've excluded 2; can you make your bounds more precise given this information?

3 Asymptotics

So far our applications of size methods have been pretty tame. We've mostly considered bounding applications in the context of factoring and divisibility estimates. However we can push size methods much further, by noting that using some *analysis* (in the calculus sense), we may estimate the size of many functions in various limits (notably as our parameters go to infinity). This can be a very good tool for both developing preliminary intuition on a problem and also performing concrete estimates.

Here is my favourite example of this idea:

Example 3.1. [CMO 2020 4] Let $S = \{1, 4, 8, 9, 16, \dots\} := \{n^k | n, k \in \mathbb{Z}, k \geq 2\}$ denote the set of perfect integer powers. Arrange the elements in S into an increasing sequence $\{a_i\}$. Show that there exist infinitely many positive integers n satisfying $9999 | a_{n+1} - a_n$.

To start out, let's think about some meta-commentary. What do we know about S ? Perfect powers are somewhat easy to work with on their own, and if we just restricted to a specific type of perfect power (e.g. only perfect cubes), then it would be easy to understand how this consecutive ordering works. But putting all perfect powers together muddies this ordering a lot. How do we deal with that?

1. Let S_k be the set of perfect k^{th} powers, arranged in increasing order. Can you prove the problem if S is replaced with S_k for some k of your choice?

2. Ok, let's now think about how to work with S . The key is that *some perfect powers are much more common than others*. Specifically, in your mind, think about:
 - (a) In your chosen S_k , how often does $9999|a_{n+1} - a_n$ occur?
 - (b) How often do non- k^{th} powers appear in S ?
3. Make the above estimates rigorous and solve the problem.

Here's another example where you may have to think more carefully about how to apply your asymptotic methods.

Example 3.2. [EGMO 2021 6] Does there exist a nonnegative integer a for which the equation

$$\left\lfloor \frac{m}{1} \right\rfloor + \left\lfloor \frac{m}{2} \right\rfloor + \left\lfloor \frac{m}{3} \right\rfloor + \cdots + \left\lfloor \frac{m}{m} \right\rfloor = n^2 + a$$

has more than one million different solutions (m, n) where m and n are positive integers?

1. First, do a rough estimate of the LHS, and think about whether you think the answer is “yes” or “no”. (You don't have to make a decision yet!)
2. Let's try to formalize a way to count how often an individual value for a “receives” a working pair (m, n) . Denote the above left-hand-side by $f(m)$. Pick a large number K , and estimate, for each n , how many values of m satisfy $f(m) - n^2 \in [0, K]$.
3. Show that when summed over all n , we can choose K to be large enough such that for any constant c , our count above exceeds cK . Using this, solve the problem.

In fact, the estimate in b) above does not need to be very good at all.

In future examples you do, it may be useful to know some estimates of classical number-theoretic functions. Here are two important facts:

1. Bertrand's Postulate: for every positive integer $n > 1$, there exists a prime number p satisfying $n < p < 2n$.
2. Prime Number Theorem: for any positive real x , let $\pi(x)$ denote the number of prime numbers not greater than x . Then

$$\lim_{x \rightarrow \infty} \frac{\pi(x)}{x / \ln x} = 1$$

4 Summary

Here are some ideas we learned today:

1. We learned some tools for dealing with integer divisibility. Notably we learned how Euclidean Algorithm-style bounding can be a useful technique for attacking such problems.
2. We learned how to analyze equalities through the lens of inequalities, by carefully inspecting the sizes of our terms. In an NT context, we found it useful to factor our equations when doing this.
3. We learned that understanding expressions via their limits at infinity can be useful conceptually and practically.

5 Problems

Below, the diamond $\diamond$ problems are ones which are in the class version of the handout, and are problems I have more info (e.g. hints) on. I think the starred $\star$ problems are very nice and you should do them if possible.

5.1 Part A

Here are some IMO 1+ level problems.

1. $\diamond$ Consider a collection of real numbers $a_1, \dots, a_k$, not all of which are integers. Prove that for infinitely many integers n , the integers n and $\lfloor na_1 \rfloor + \dots + \lfloor na_k \rfloor$ are relatively prime.
2. $\diamond$ Let $n \geq 2$ be a positive integer, with divisors $1 = d_1 < d_2 < \dots < d_k = n$. Let $D = d_1 d_2 + d_2 d_3 + \dots + d_{k-1} d_k$.
 - (a) Prove that D is always less than n^2 .
 - (b) Determine when D is a divisor of n^2 .

3. For any positive integer k , denote the sum of digits of k in its decimal representation by $S(k)$. Find all polynomials $P(x)$ with integer coefficients such that for any positive integer $n \geq 2016$, the integer $P(n)$ is positive and

$$S(P(n)) = P(S(n)).$$

4. $\diamond$ Let $a_1, \dots, a_n, b_1, \dots, b_n$ be $2n$ positive integers such that the $n+1$ products

$$a_1 a_2 a_3 \cdots a_n, b_1 a_2 a_3 \cdots a_n, b_1 b_2 a_3 \cdots a_n, \dots, b_1 b_2 b_3 \cdots b_n$$

form a strictly increasing arithmetic progression in that order. Determine the smallest possible integer that could be the common difference of such an arithmetic progression.

5. $\star$ Find all positive integers n with the following property: the k positive divisors of n have a permutation $(d_1, d_2, \dots, d_k)$ such that for $i = 1, 2, \dots, k$, the number $d_1 + d_2 + \dots + d_i$ is a perfect square.
6. $\diamond$ Let a and b be positive integers such that $a! + b!$ divides $a!b!$. Prove that $3a \geq 2b + 2$.
7. $\diamond$ Consider a polynomial $P(x) = \prod_{j=1}^9 (x + d_j)$, where $d_1, d_2, \dots, d_9$ are nine distinct integers. Prove that there exists an integer N , such that for all integers $x \geq N$ the number $P(x)$ is divisible by a prime number greater than 20.
8. $\star$ Let $P(x)$ be a monic integer polynomial of even degree. Then $P(n)$ is a perfect square for infinitely many integers n iff P itself is the square of a polynomial.

5.2 Part B

Here are some IMO 2+ level problems.

1. Determine all sequences $(x_1, x_2, \dots, x_{2011})$ of positive integers, such that for every positive integer n there exists an integer a with

$$\sum_{j=1}^{2011} j x_j^n = a^{n+1} + 1$$

2. $\diamond$ An infinite sequence of positive integers $a_1, a_2, \dots$ is called *good* if (1) a_1 is a perfect square, and (2) for any integer $n \geq 2$, a_n is the smallest positive integer such that

$$n a_1 + (n-1) a_2 + \dots + 2 a_{n-1} + a_n$$

is a perfect square. Prove that for any good sequence $a_1, a_2, \dots$, there exists a positive integer k such that $a_n = a_k$ for all integers $n \geq k$.

3. $\diamond$ Let $a_1, a_2, a_3, \dots$ be a sequence of positive integers and let $b_1, b_2, b_3, \dots$ be the sequence of real numbers given by

$$b_n = \frac{a_1 a_2 \cdots a_n}{a_1 + a_2 + \cdots + a_n}, \text{ for } n \geq 1$$

Show that, if there exists at least one term among every million consecutive terms of the sequence $b_1, b_2, b_3, \dots$ that is an integer, then there exists some k such that $b_k > 2021^{2021}$.

4. $\star$ Let n be a positive integer. Let D_n be the set of all divisors of n and let $f(n)$ denote the smallest natural m such that the elements of D_n are pairwise distinct in mod m . Show that there exists a natural N such that for all $n \geq N$, one has $f(n) \leq n^{0.01}$.
5. Prove that for every positive integer n , there exists a positive integer k such that

$$a_1! + a_2! + \cdots + a_n! + k$$

is not a perfect square for any positive integers $a_1, a_2, \dots, a_n$.

6. $\diamond$ For a polynomial P and a positive integer n , define P_n as the number of positive integer pairs (a, b) such that $a < b \leq n$ and $|P(a)| - |P(b)|$ is divisible by n . Determine all polynomial P with integer coefficients such that $P_n \leq 2021$ for all positive integers n .
7. Let n, m be integers greater than 1, and let $a_1, a_2, \dots, a_m$ be positive integers not greater than n^m . Prove that there exist positive integers $b_1, b_2, \dots, b_m$ not greater than n , such that

$$\gcd(a_1 + b_1, a_2 + b_2, \dots, a_m + b_m) < n,$$

where $\gcd(x_1, x_2, \dots, x_m)$ denotes the greatest common divisor of $x_1, x_2, \dots, x_m$.

8. Let $\mathbb{Z}$ denote the set of all integers. Find all polynomials $P(x)$ with integer coefficients that satisfy the following property:
For any infinite sequence $a_1, a_2, \dots$ of integers in which each integer in $\mathbb{Z}$ appears exactly once, there exist indices $i < j$ and an integer k such that $a_i + a_{i+1} + \cdots + a_j = P(k)$.
9. $\star$ Prove that there is a strictly increasing positive integer sequence (a_n) and a constant $K > 0$, such that $a_n \leq 1.01^n K$ for all n and the sum of any finite number of distinct terms in (a_n) is not a perfect square.
10. $\star$ Let a, b, c, d, e be distinct positive integers such that $a^4 + b^4 = c^4 + d^4 = e^5$. Show that $ac + bd$ is a composite number.

5.3 Part C

Here are some IMO 3+ level problems.

1. $\diamond$ Let $m > 1$ be an integer. A sequence $a_1, a_2, a_3, \dots$ is defined by $a_1 = a_2 = 1$, $a_3 = 4$, and for all $n \geq 4$,

$$a_n = m(a_{n-1} + a_{n-2}) - a_{n-3}.$$

Determine all integers m such that every term of the sequence is a square.

2. $\diamond$ Let K be the set of all positive integers that do not contain the digit 7 in their base-10 representation. Find all polynomials f with nonnegative integer coefficients such that $f(n) \in K$ whenever $n \in K$.
3. $\star$ Prove that there exist infinitely many integers n such that $n^2 + 1$ is squarefree.
4. Let $\mathbb{N}$ be the set of positive integers. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function satisfying the following two conditions:
(a) $f(m)$ and $f(n)$ are relatively prime whenever m and n are relatively prime. (b) $n \leq f(n) \leq n + 2012$ for all n .

Prove that for any natural number n and any prime p , if p divides $f(n)$ then p divides n .

5. Prove that there is a constant $c > 0$ with the following property: If a, b, n are positive integers such that $\gcd(a + i, b + j) > 1$ for all $i, j \in \{0, 1, \dots, n\}$, then

$$\min\{a, b\} > c^n \cdot n^{\frac{n}{2}}.$$

5.3.1 Advanced Asymptotic Party

Here are some advanced NT problems which utilize size methods. Many of these problems come from Chinese sources (e.g. China TST), which you can check out. I haven't really done these but I would be willing to work on these with you.

1. Let $a_1, a_2, a_3, \dots$ be distinct positive integers, and $0 < c < \frac{3}{2}$. Prove that there exist infinitely many positive integers k , such that $\gcd(a_k, a_{k+1}) > ck$.
2. Prove that for any odd prime number p , the number of positive integer n satisfying $p|n! + 1$ is less than or equal to $cp^{\frac{2}{3}}$, where c is a constant independent of p .
3. Let $b_1 < b_2 < b_3 < \dots$ be the sequence of all natural numbers which are sum of squares of two natural numbers. Prove that there exists infinite natural numbers like m which $b_{m+1} - b_m = 2015$.
4. Let $a_1, a_2, \dots, a_{2^{2016}}$ be positive integers not larger than 2016. We know that for each $n \leq 2^{2016}$, $a_1 a_2 \dots a_n + 1$ is a perfect square. Prove that for some i , $a_i = 1$.
5. Prove that there is a constant $c > 0$ and infinitely many positive integers n with the following property: there are infinitely many positive integers that cannot be expressed as the sum of fewer than $cn \log(n)$ pairwise coprime n th powers.
6. Let $a_1, a_2, \dots, a_n$ be n positive integers that are not divisible by each other, i.e. for any $i \neq j$, a_i is not divisible by a_j . Show that

$$a_1 + a_2 + \dots + a_n \geq 1.1n^2 - 2n.$$